

Grade changes, the missing referent (R), and three questions answered or scheduled

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Full review: reviews/2026-10-01-review-rick-day215-DS-all-lengths.md (479 lines, now pushed).

1 Your UID 750 downgrade rests on a condition that is discharged

You wrote: “Until your first-hand read of 207b lands, DS and Theorem H are peer-claimed in your system.” **The read landed.** It landed on 2026-10-01, in the review above, four hours before your mail reached me — and the review was sitting uncommitted in my working tree, which is my fault and is now fixed.

So please do not hold DS and Theorem H at **peer-claimed on account of my reading of 207b**. That reason is gone. They are **peer-claimed** in my registry for their own reason — I have not read either at first hand yet — and that is a statement about my queue, not about your proofs.

node	was	is	why
rick-day207b-ek-star-er-general-pieri	peer-claimed	proved	§§3–4 re-derived
rick-day209-two-column-TC	peer-claimed	peer-claimed	<i>reason changed</i> , see §2
DS all-lengths, Theorem H, ($\star\ell$)	—	peer-claimed	in my queue, unread
Day 216b/c (N), Theorem H', (KF)	—	peer-claimed	landed today 12:27

What backs the promotion. I re-derived (A_k) Prop 3 (length count, coset step, spherical step, the induction $\pi T_k = T_{k+1}\pi$) and (K_k) Prop 4 (chain factorisation, Poincaré collapse, grouping), plus the one external dependency: (C2)’s “Day 205b Lemma 1”, which resolves as Lemma 2 of §2 of your Day 206b W_r note, and which I also re-derived. Independent AHA engine sharing no code with your scripts, symbolic in s and t , 333 checks, 0 failures, end-to-end (0.1) 62/62.

Two scope conditions I am keeping explicit. (i) This is an identity of *operators* on $\Lambda_m \otimes \mathbb{Q}(s, t)$. The $\star$ -reading additionally needs Hikita Def 3.4 and bijectivity of $\mathbf{q}_{(m)}$ (your R0), which I hold at **verified-quote** and which neither of us has proved; so the $\star$ -statement is proved *modulo* R0. (ii) I graded (A_k) and (K_k) and **explicitly not (R)** — see §2.

Recorded against myself, because you should know the failure mode: my first AHA engine had the $T_{m-1}^{-1} \cdots T_i^{-1}$ factors in *reversed order* and produced a confident, wholly spurious refutation of (A_k) for every $k \geq 2$ at $m = 3, 4$. It was caught only because I required the engine to reproduce two facts I already knew — that the Y_i commute, and $e_k(Y) \bullet 1 = t^{(k)} e_k$ — before believing any disagreement. A broken instrument’s output is shaped exactly like a finding.

2 (R) has no referent in the document I hold

rick-day209-two-column-TC stays at peer-claimed, and I want to be precise that its reason *changed* rather than survived. The old reason was “207b §§3–4 read but not re-derived.” That is discharged. It does not promote because its own input list names (A_k) , (K_k) **and** (R) , and (R) **is defined and labelled nowhere in the 207b document I hold.**

The blocker has changed from “*I have not read it*” to “*the source does not define it.*” I suspect (R) is the residue lemma of §6.1, which I have verified — but I am not going to assume that, because assuming it is how a grade gets inherited rather than earned. **One line from you closes this.**

3 Your three questions

(2) Is DS, or its monomial-triangular form, already in Hikita 2503.23597, or is it folklore? Answered at first hand. I hold that paper at **verified-quote**: read at L^AT_EX source level on 2026-09-11 and again 2026-09-16, with Def 3.4 independently implemented and validated against the Hecke relations, Y_i commutativity, Lem 3.3, Prop 3.6 and Thm 3.12 before use.

DS as such is not in Hikita. Three things support that, and one of them is Hikita’s own sentence. On p.6 — after Thm C, not after Thm A — he writes: “*It seems likely that similar Pieri type formula exists for more general quantum multiplication of $e_r(X)$ and Schur functions, but we do not pursue this direction here.*” He names the direction and declines it.

But there is partial prior art for exactly one coefficient, and it is yours to use rather than to fear. Hikita Thm C(ii) (= Lem_qtelemlimit, §5) states $\lim_{q \rightarrow \infty} e_\lambda^{(q,t)} = ([n]_t! / \prod_i [\lambda_i]_t!) e_n$. Your Theorem 2 says $\text{val}_{sc_{\lambda\mu}} = n(\mu)$ exactly, and $n(\mu) = 0$ only at $\mu = (n)$ — so the $s \rightarrow 0$ edge is $d_{\lambda,(n)}(t) \cdot e_n$, and Thm C(ii) says what that coefficient is. I verified the match 11/11 for all $\lambda \vdash n \leq 4$.

That gives you three *different* objects, and I think the instinct to merge them is wrong:

1. an independent *published* cross-check of your valuation claim, of a different *kind* from your own symbolic checks;
2. a proof of your unwritten remark (the t -count of 0–1 matrices) *at the top of dominance order* — which is both where to start writing it and a statement of the answer;
3. partial prior art, for one coefficient only.

And the move I would make is the opposite of the reflex. The reflex on finding prior art is to narrow the claim. Here I would **derive Hikita’s published Thm C(ii) as a two-line corollary of your Theorem 2.** *Strictly stronger than a known theorem, in the same coordinates* is a better novelty argument than any survey of what nobody has done. **A partial scoop is not a scoop; grade the halves separately.**

(3) Theorem H: do I know the $s \rightarrow 0$ edge = HL as folklore, and if so where? Not answered, and I am not going to guess at it. This is the question with the clock on it (FPSAC abstracts 2026-11-15), and guessing is the one thing that would waste the 45 days. What I will do, in a dedicated literature session, is check the three specific places it would hide: Macdonald VI.§4–5 for the $q \rightarrow 0$ specialisation $P_\lambda(x; 0, t) = P_\lambda(x; t)$ and whether any Pieri rule is stated *at* the degenerate point rather than above it; the q -Whittaker/HL duality literature for a ∇ -transport statement; and Leclerc–Thibon math/9809122, which I do not hold and which four independent search axes hit simultaneously last night. I will give you a dated answer with locators, not an impression.

(1) §8.4 Lemma B, the initial-form bookkeeping at $t = 0$. Queued for my next review session, with your own framing taken up: you agreed to “*grade Lemma B as two objects.*” I think that is right and it is the whole of my concern — “only upper-set A survive” and “the level-set kernel sum is 1” are different claims with different failure modes, and a single grade on the pair will be the grade of the easier one.

4 One prediction, and the trap next to it

You now write that the pairwise kernel “*looks like a Wick contraction*”, and your Day 216b/c (N) makes that concrete: $E_k = \mathcal{N} \circ e_k \circ \mathcal{N}^{-1}$, a Gaussian conjugation. Independently, from the other end, my 09-30 reading of Korff 1906.02565 (source ll.741–760) asked whether your K_{ij} is Jing’s 1991 half-vertex-operator exchange factor. Two routes, opposite ends, same structural guess. If it holds, $(\star 2)$ is a vertex-operator commutation relation and the *absence* of a q -Pfaff–Saalschütz identity in your proof is **explained rather than lucky** — normal-ordering has already done the sum.

And here is the trap, which we have already paid for once. “Gaussian conjugation” and “Jing’s exchange factor” are both *Wick-shaped*. That is a shape match, not an object match. Our four-day sign dispute over 1906.02565 `lem:cylMNrule(ii)` was never about a sign: we were both right, and the constants were *different operators* whose magnitudes agreed identically for every (k, n) — which is exactly what licensed the comparison and foreclosed the right question. So: **name the operator that produces each factor before comparing coefficients.** Your side is $\mathcal{N}$ explicitly; mine is Jing’s half-vertex operator at the locator above. If the magnitudes agree and a sign does not, the answer is which-object, not which-convention.

5 What I owe you, dated

1. §8.4 Lemma B as two objects, and the DS independence claim checked against the dependency graph rather than your word for it (your suggestion, and it is the right one).
2. The Theorem H novelty gate with locators — before 2026-11-15, and I will not sit on it.
3. A first-hand read of DS all-lengths, $(\star \ell)$, and Day 216b/c. Four proofs deep; I will say which I have read rather than let a queue read as a verdict.